

ECB1323 ANTENNA AND MILLIMETRE WAVE COMMUNICATION

DESIGN OF T-SHAPED MICROSTRIP ANTENNA FOR RADAR COMMUNICATION APPLICATION

A PROJECT REPORT

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(Autonomous Institution affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report “**DESIGN OF T-SHAPED MICROSTRIP ANTENNA FOR RADAR COMMUNICATION APPLICATION** ” is the bonafide work of “**KAMALESH R (927623BEC099) , KAVIN R K (927623BEC109) , NAVEENKUMAR S (927623BEC304) , SARATHI M (927623BEC401)** ” who carried out the project work during the academic year 2025- 2026 under my supervision.

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ABSTRACT

Microstrip patch antennas are widely used in modern radar communication systems due to their compact structure, low fabrication cost, and ease of integration with microwave circuits. With the increasing demand for high-frequency sensing applications, antennas operating in the millimeter-wave region have gained significant importance. In this report, a T-shaped microstrip patch antenna operating at 26 GHz is designed and analyzed for radar communication applications. The proposed antenna structure improves impedance matching, bandwidth performance, and radiation characteristics compared to conventional rectangular microstrip antennas by increasing the effective current path length on the radiating patch. The antenna geometry is designed using standard transmission-line model equations and simulated using ANSYS HFSS electromagnetic simulation software. Performance parameters such as return loss (S_{11}), voltage standing wave ratio (VSWR), gain, and radiation pattern are evaluated to verify antenna characteristics at the desired operating frequency. The simulated results indicate satisfactory radiation performance with reduced reflection losses and stable directional characteristics, making the proposed antenna suitable for short-range radar sensing, automotive radar systems, and millimeter-wave communication applications.

Abstract (Key words)	POs Mapping
Microstrip Patch Antenna, T-Shaped Antenna, Radar Communication Systems, Millimeter-Wave Antenna, Return Loss (S11), VSWR, Radiation Pattern, Gain, HFSS Simulation, Rogers RO4003 Substrate, Directional Antenna Design, High-Frequency Antenna	PO1, PO2, PO3, PO4, PO5, PO7, PO10, PSO1, PSO2

SDG Goal		Remarks
SDG 11 	Enables antenna structures suitable for smart transportation radar sensing applications.	Improves safety in intelligent vehicular communication environments.

Project Component	Relevant IEEE Standards
Antenna measurement procedures	IEEE 149 – Standard Test Procedures for Antennas
Radar frequency band classification	IEEE 521 – Radar Frequency Band Designations
Electromagnetic simulation validation	IEEE 1597 – Computational Electromagnetics Modeling
Antenna parameter definitions	IEEE 145 – Definitions of Terms for Antennas
RF impedance and transmission measurements	IEEE 1785 – Transmission Line Measurement Standards
High-frequency wireless compatibility	IEEE 802.15 – Wireless Communication Standards

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LIST OF ABBREVIATIONS

ACRONYM	ABBREVIATION
RF	Radio Frequency
EM	Electromagnetic
GHz	Gigahertz
MSA	Microstrip Antenna
TMPA	T-Shaped Microstrip Patch Antenna
DGS	Defected Ground Structure
HFSS	High Frequency Structure Simulator
S11	Reflection Coefficient
VSWR	Voltage Standing Wave Ratio
RL	Return Loss

CHAPTER 1

INTRODUCTION

1.1 RADAR COMMUNICATION SYSTEMS

1.1.1 Definition of Radar

Radar, an acronym for *Radio Detection and Ranging*, is an electromagnetic sensing technique used to determine the presence, position, distance, direction, and velocity of objects by transmitting high-frequency electromagnetic waves toward a target and analyzing the signals reflected from it. The operation of radar systems is based on the principle that electromagnetic waves propagate through free space at the speed of light and undergo reflection when they encounter discontinuities such as aircraft, vehicles, terrain surfaces, or atmospheric disturbances. Radar systems are capable of operating effectively under adverse environmental conditions such as fog, rain, dust, smoke, and darkness, where optical sensing techniques become unreliable. Because of these advantages, radar communication systems have become indispensable in modern engineering applications including air traffic control, marine navigation, weather monitoring, military surveillance, and automotive safety systems.

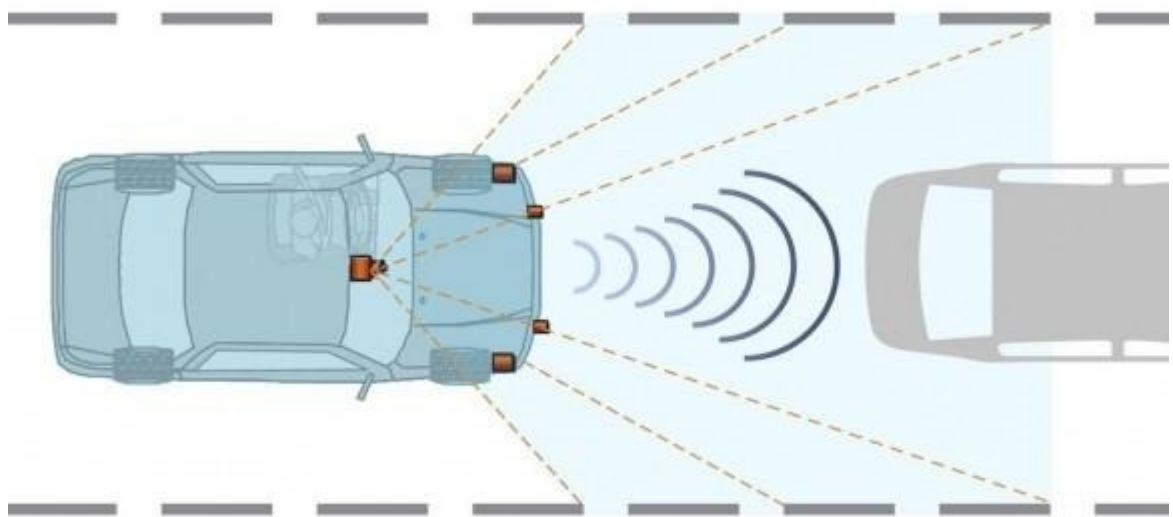


Fig 1.1 Radar Communication

1.1.2 Working Principle of Radar

The working principle of radar systems involves the transmission of electromagnetic energy from a transmitting antenna toward a target object and the reception of the reflected signal by a receiving antenna. When an electromagnetic wave encounters an object, a portion of the incident energy is scattered in various directions, and part of this scattered energy returns toward the radar receiver. The distance between the radar system and the target can be determined by measuring the time delay between transmission and reception of the signal, which is expressed as

$$R = \frac{ct}{2} \quad \text{----- 1.1}$$

where R represents the distance between the radar and the target, c represents the velocity of electromagnetic waves in free space, and t represents the round-trip travel time of the signal. In addition to range measurement, radar systems can determine target velocity using the Doppler frequency shift principle, which arises due to relative motion between the radar system and the target. These capabilities enable radar systems to perform accurate detection, tracking, and monitoring of stationary as well as moving objects.

1.1.3 Applications of Radar Communication Systems

Radar communication systems are widely used in several civilian and defense applications because of their ability to provide reliable object detection independent of environmental visibility conditions. In automotive engineering, radar sensors operating in the microwave and millimeter-wave frequency bands are extensively used for adaptive cruise control, collision avoidance detection, blind-spot monitoring, and lane-change assistance systems. In military applications, radar systems are used for long-range surveillance, missile guidance, target tracking, and reconnaissance operations. Weather monitoring

systems utilize radar technology to measure precipitation intensity, storm movement, and atmospheric disturbances.

1.2 ELECTROMAGNETIC SPECTRUM

1.2.1 Overview of Electromagnetic Spectrum

The electromagnetic spectrum represents the complete range of electromagnetic radiation arranged according to frequency or wavelength. Electromagnetic waves are generated by the acceleration of electric charges and propagate through free space in the form of mutually perpendicular electric and magnetic fields. These waves travel at the speed of light, which is approximately $3 \times 10^8 m/s$ in free space. The relationship between frequency and wavelength of an electromagnetic wave is expressed as

$$\lambda = \frac{c}{f} \quad \text{----- 1.2}$$

where λ represents wavelength, c represents the velocity of light, and f represents frequency of operation.

The electromagnetic spectrum extends from extremely low-frequency waves to very high-frequency gamma rays. Each portion of the spectrum exhibits unique propagation characteristics that determine its suitability for different communication, sensing, imaging, and scientific applications. The lower frequency regions of the spectrum are primarily used for long-distance communication systems, while higher frequency regions such as microwave and millimeter-wave bands are widely used in radar systems, satellite communication, and high-speed wireless communication technologies. As frequency increases, wavelength decreases, resulting in smaller antenna dimensions and improved resolution in sensing applications.

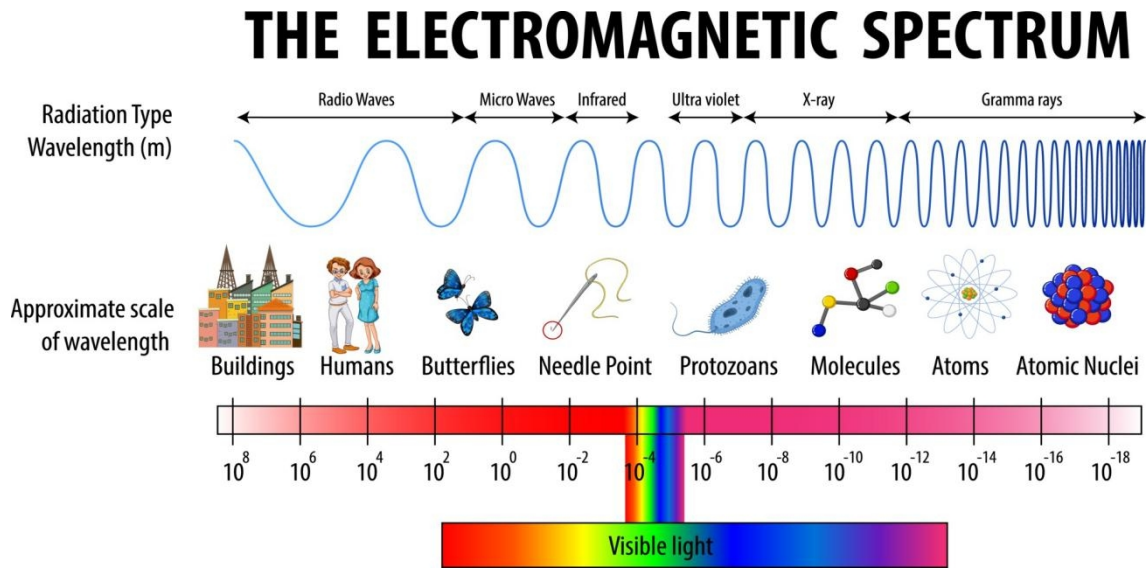


Fig 1.2 EM Spectrum

1.2.2 Microwave Frequency Band

Microwave frequency bands are widely used in radar communication systems, satellite communication systems, wireless networking applications, and remote sensing technologies. These frequencies provide moderate bandwidth availability and stable propagation characteristics suitable for medium-range communication systems. Antennas designed at microwave frequencies generally require larger physical dimensions compared to antennas operating in millimeter-wave frequency ranges. However, operation near the upper microwave region around 26 GHz enables reduction in antenna size while maintaining efficient radiation performance. This makes the 26 GHz frequency band particularly suitable for compact radar sensing applications requiring high spatial resolution.

1.2.3 Millimeter-Wave Frequency Band

Millimeter-wave frequencies extend from 30 GHz to 300 GHz and provide significantly wider bandwidth availability compared to conventional microwave frequency bands. Although the proposed antenna operates slightly below this region, it shares several advantages associated with millimeter-wave communication systems such as reduced antenna dimensions, improved

directivity, and enhanced detection accuracy. Millimeter-wave antennas are widely used in automotive radar systems, 5G wireless communication technologies, satellite communication systems, and high-speed wireless data transmission applications where compact size and high performance are essential.

1.3 RADIO FREQUENCY SYSTEMS AND RF DEVICES

1.3.1 Introduction to RF Systems

Radio frequency communication systems operate by transmitting electromagnetic signals through free space within the frequency range extending approximately from 3 kHz to 300 GHz. These systems form the foundation of modern wireless communication technologies including radar communication systems, satellite communication systems, broadcasting systems, and mobile communication networks. In radar communication applications, RF signals generated by oscillators are amplified and transmitted through antennas toward the target object, and the reflected signals are received and processed to determine target location and velocity. Operation at higher RF frequencies such as 26 GHz enables compact antenna structures with improved spatial resolution and enhanced directional radiation characteristics, making them highly suitable for modern radar sensing applications.

1.3.2 Components of RF Communication Systems

An RF communication system consists of several essential components responsible for signal generation, processing, amplification, filtering, and transmission. Oscillators generate carrier signals required for communication, while amplifiers increase signal strength prior to transmission. Mixers perform frequency conversion during modulation and demodulation processes, and filters remove unwanted frequency components from transmitted and received signals. Transmission lines guide electromagnetic energy between circuit components with minimal losses. Among these components, antennas play the most important role because they convert guided electromagnetic energy into radiated

waves and determine communication range, radiation efficiency, and signal quality

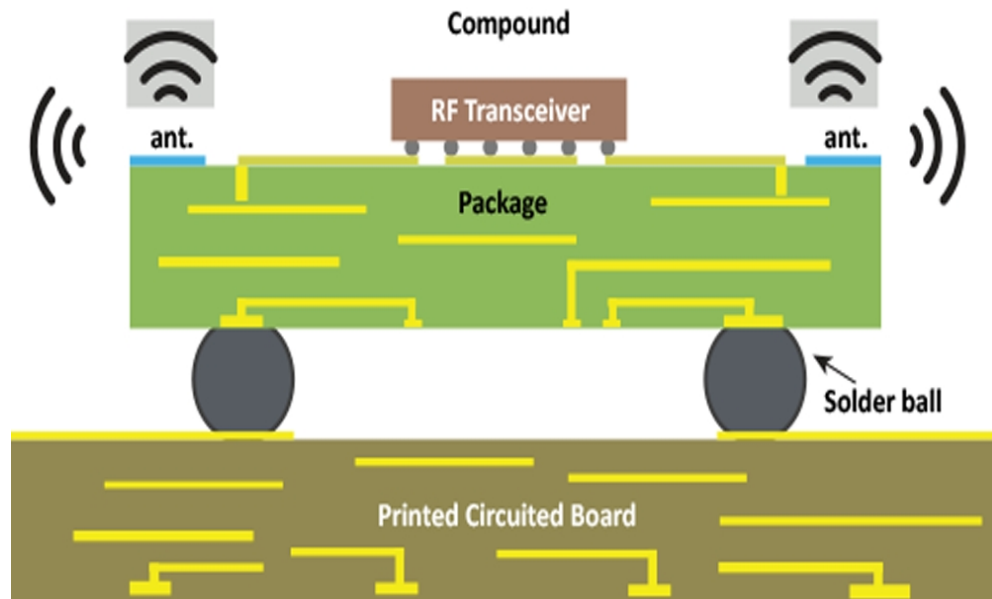


Fig 1.3 RF System

1.3.3 Role of RF Antennas in Radar Systems

In radar communication systems, antennas function as both transmitting and receiving elements that convert electrical RF signals into electromagnetic waves and vice versa. During transmission, the antenna radiates electromagnetic energy toward the target object, and during reception it captures reflected signals returning from the target. The performance of radar systems depends strongly on antenna parameters such as gain, bandwidth, radiation pattern, and impedance matching characteristics. Efficient antenna design therefore improves detection accuracy, signal resolution, and overall system performance.

1.4 ANTENNA

1.4.1 Definition of Antenna

An antenna is a transducing device that converts electrical energy into electromagnetic radiation during transmission and converts electromagnetic radiation into electrical signals during reception. It acts as an interface between guided electromagnetic waves travelling through transmission lines and unguided waves propagating through free space. When an alternating current flows through a conducting antenna structure, time-varying electric and magnetic fields are produced around the conductor, which detach from the antenna surface and propagate outward as electromagnetic waves. Similarly, when electromagnetic waves incident on the antenna induce currents on its surface, these currents are delivered to the receiving circuitry for signal processing. Because of this bidirectional property, antennas form essential components in all wireless communication systems including radar communication systems, satellite communication systems, broadcasting systems, navigation systems, and mobile communication networks.

1.4.2 Basic Principle of Radiation

Radiation from an antenna occurs due to the acceleration or deceleration of electric charges in a conducting medium. When time-varying current flows through an antenna conductor, oscillating electric and magnetic fields are generated around the conductor. These fields propagate away from the antenna structure in the form of electromagnetic waves travelling at the speed of light. The electric field and magnetic field components are mutually perpendicular to each other and also perpendicular to the direction of wave propagation. The relationship between wavelength and frequency of the electromagnetic wave is given by

$$\lambda = \frac{c}{f} \quad \text{----- 1.3}$$

where λ represents wavelength, c represents velocity of light, and f represents operating frequency. Efficient radiation occurs when the physical length of the antenna is comparable to the wavelength of the operating signal.

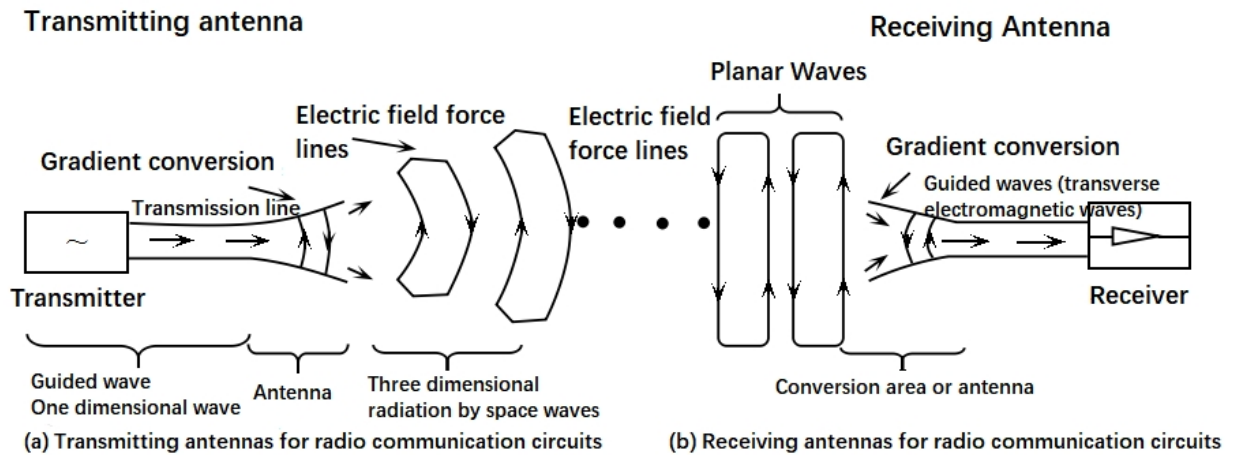


Fig 1.4 Antenna working

1.4.3 Antenna as a Transmission System Element

In a wireless communication system, the antenna acts as a transition device between transmitter and receiver through the free-space communication channel. During transmission, electrical energy generated by the transmitter is guided through a transmission line and supplied to the antenna, where it is converted into electromagnetic radiation and propagated into space. During reception, the antenna intercepts electromagnetic waves travelling through the communication channel and converts them back into electrical signals that are delivered to the receiver. The efficiency with which this conversion process takes place determines the overall performance of the communication system. Therefore, proper antenna design plays a crucial role in achieving reliable signal transmission and reception.

1.4.4 Antenna Radiation Regions

The space surrounding an antenna is generally divided into three regions known as the reactive near-field region, radiating near-field region, and far-field region. In the reactive near-field region, electromagnetic energy is mainly stored rather than radiated and strong reactive fields exist close to the antenna surface. In the radiating near-field region, the radiation pattern depends on the distance from the antenna and begins to stabilize gradually. In the far-field region, electromagnetic waves propagate freely with well-defined radiation characteristics independent of distance from the antenna. Most antenna parameters such as gain, directivity, and radiation pattern are measured in the far-field region because the radiation becomes uniform and predictable in this region.

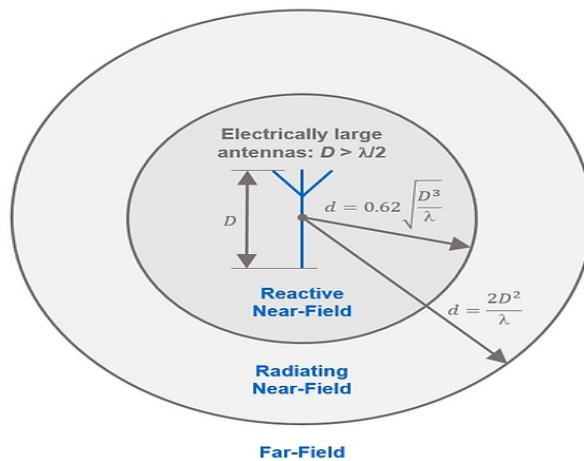


Fig 1.5 Radiation region

1.4.5 Classification of Antennas

Antennas are classified into different categories depending on their structure, operating frequency, and radiation characteristics. Wire antennas such as dipole antennas and monopole antennas are among the simplest antenna types and are widely used in broadcasting systems. Aperture antennas such as horn antennas are commonly used in microwave communication systems due to their

high directivity and wide bandwidth characteristics. Reflector antennas such as parabolic reflector antennas are widely used in satellite communication systems and long-range radar applications where high gain is required. Array antennas consist of multiple radiating elements arranged in a specific geometrical configuration to achieve improved radiation control and enhanced directivity. Printed antennas represent another important class of antennas that are widely used in compact wireless communication systems due to their planar structure and ease of fabrication.

Different Types of Antennas

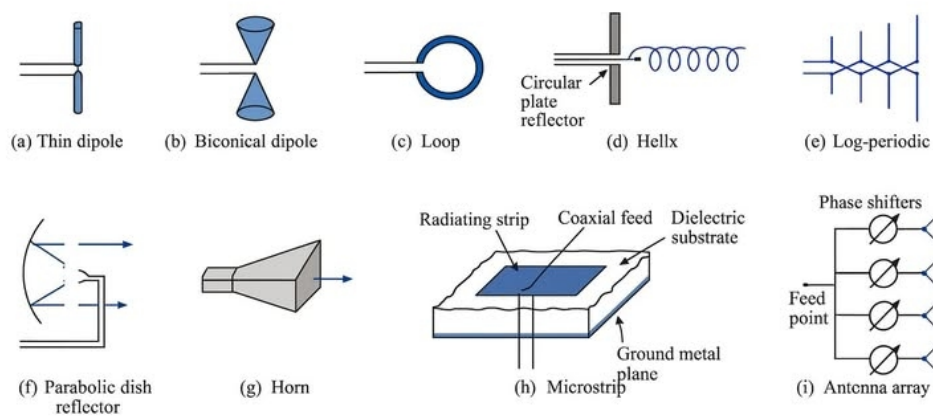


Fig 1.6 Types of antennas

1.4.6 Antenna Polarization

Polarization of an antenna describes the orientation of the electric field vector of the radiated electromagnetic wave with respect to the direction of propagation. Depending on the orientation of the electric field, polarization may be classified as linear polarization, circular polarization, or elliptical polarization. In linear polarization, the electric field vector remains confined to a single plane during wave propagation.

1.5 ANTENNA PARAMETERS

1.5.1 Radiation Pattern

The radiation pattern of an antenna represents the spatial distribution of radiated electromagnetic energy as a function of direction in space. It describes how power is radiated from an antenna in different directions and provides important information regarding directional characteristics of the antenna. Radiation patterns are generally represented in two-dimensional polar plots or three-dimensional graphical formats, where the variation of radiation intensity is observed with respect to angular position.

The radiation intensity is expressed as

$$U = \frac{P_{rad}}{4\pi} \quad \text{----- 1.4}$$

where U represents radiation intensity and P_{rad} represents total radiated power.

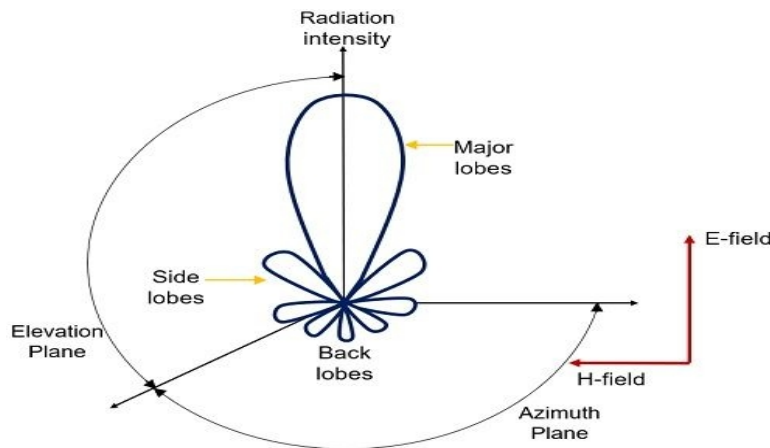


Fig 1.7 Radiation pattern

1.5.2 Gain

Gain is an important parameter that describes the ability of an antenna to concentrate radiated power in a specific direction compared to an isotropic radiator. It represents the combined effect of antenna directivity and radiation efficiency and indicates how effectively an antenna converts input power into useful radiation in a given direction. Gain is mathematically expressed as

$$G = \eta D \quad \text{----- 1.5}$$

where G represents antenna gain, η represents radiation efficiency, and D represents directivity.

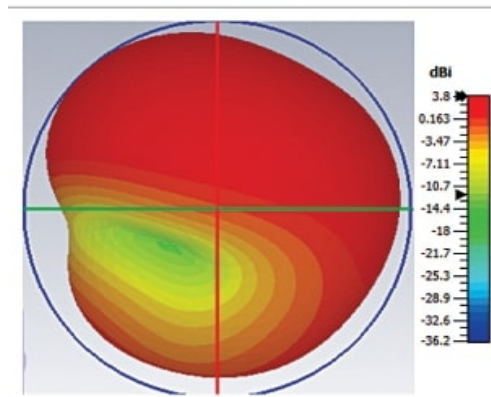


Fig 1.8 Gain Plot

Gain is commonly expressed in decibels as

$$G_{dB} = 10 \log_{10} (G) \quad \text{----- 1.6}$$

Higher gain antennas provide stronger signal transmission and reception capability, which is especially important in radar communication systems requiring improved detection range.

1.5.3 Directivity

Directivity is defined as the ratio of maximum radiation intensity in a particular direction to the average radiation intensity over all directions. It indicates how effectively an antenna focuses electromagnetic energy in a desired direction without considering internal losses.

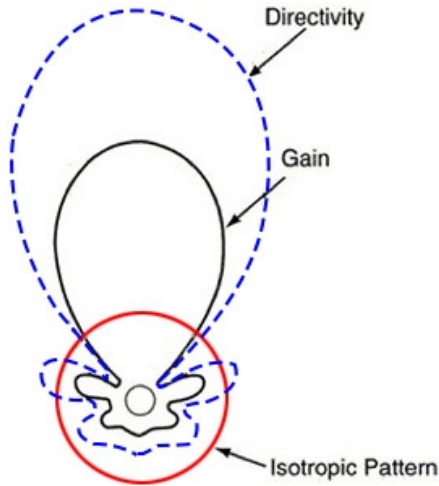


Fig 1.9 Directivity

Directivity is expressed as

$$D = \frac{4\pi U_{max}}{P_{rad}} \quad \text{----- 1.7}$$

where U_{max} represents maximum radiation intensity and P_{rad} represents total radiated power. Antennas with high directivity are widely used in radar communication applications because they provide improved angular resolution and enhanced signal focusing capability.

1.5.4 Return Loss

Return loss represents the amount of reflected power from the antenna input terminal due to impedance mismatch between the antenna and transmission line. It is an important parameter that determines how efficiently power is transferred from the transmission line to the antenna. Return loss is expressed as

$$RL = -20 \log_{10} |\Gamma| \quad \text{----- 1.8}$$

where Γ represents the reflection coefficient defined as

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0} \quad \text{----- 1.9}$$

in which Z_L represents load impedance and Z_0 represents characteristic impedance of the transmission line. A return loss value less than -10 dB indicates acceptable antenna performance, while lower values indicate better impedance matching.



Fig 1.10 Return loss

1.5.5 Voltage Standing Wave Ratio (VSWR)

Voltage Standing Wave Ratio (VSWR) describes the degree of impedance matching between the antenna and the transmission line. It is defined

as the ratio of maximum voltage to minimum voltage along the transmission line. VSWR is related to the reflection coefficient by the expression

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad \text{----- 1.10}$$

A VSWR value equal to 1 indicates perfect impedance matching, whereas values less than 2 are generally considered acceptable for efficient antenna operation. Proper impedance matching ensures maximum power transfer and minimum signal reflection in communication systems.

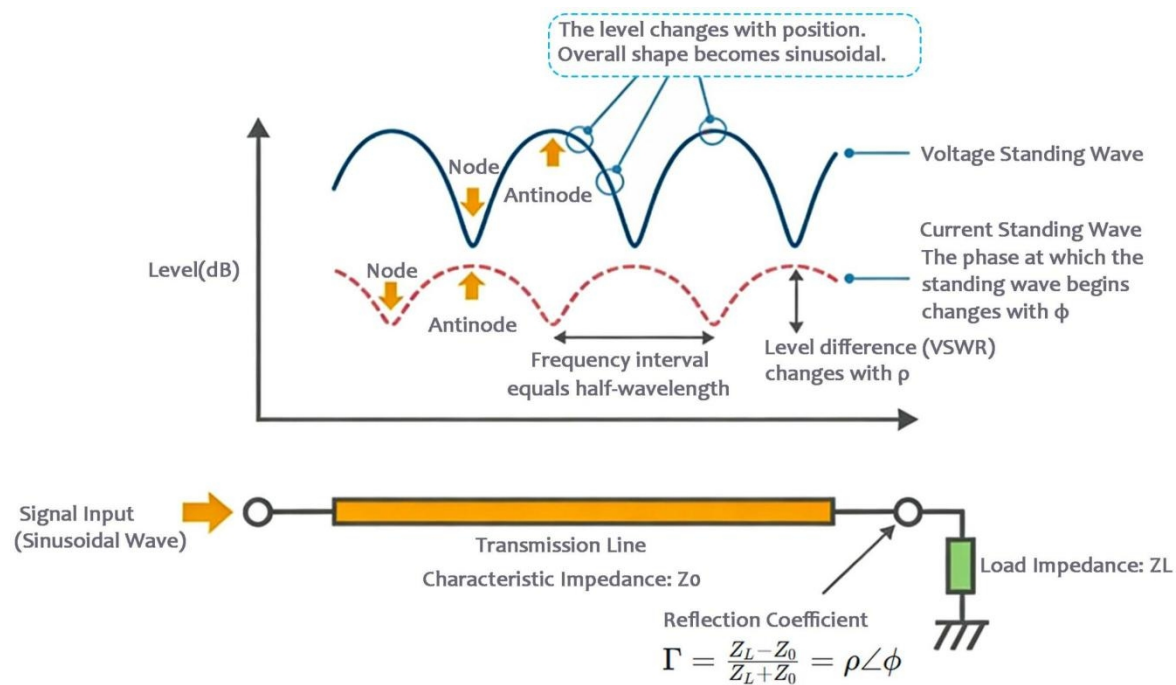


Fig 1.11 VSWR

1.5.6 Bandwidth

Bandwidth of an antenna represents the range of frequencies over which the antenna operates efficiently with acceptable performance characteristics such as gain, return loss, and radiation pattern stability. It is defined as the

difference between upper cutoff frequency and lower cutoff frequency and is expressed as

$$BW = f_H - f_L \quad \text{----- 1.11}$$

where f_H represents upper cutoff frequency and f_L represents lower cutoff frequency. Percentage bandwidth is given by

$$BW\% = \frac{f_H - f_L}{f_c} \times 100 \quad \text{----- 1.12}$$

Where f_c represents center frequency. Wide bandwidth antennas are desirable in modern communication and radar systems because they support reliable signal transmission over a broader frequency range.

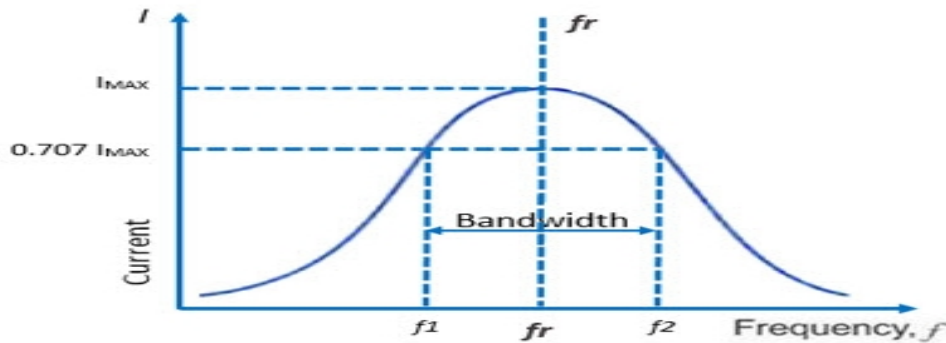


Fig 1.12 Bandwidth

1.5.7 Radiation Efficiency

Radiation efficiency represents the ratio of radiated power to total input power supplied to the antenna and indicates how effectively the antenna converts electrical energy into electromagnetic radiation. It is expressed as

$$\eta = \frac{P_{rad}}{P_{input}} \quad \text{----- 1.13}$$

where P_{rad} represents radiated power and P_{input} represents input power supplied to the antenna. Losses due to conductor resistance, dielectric absorption, and surface wave excitation reduce radiation efficiency. Higher radiation efficiency indicates better antenna performance and improved signal transmission capability.

1.5.8 Beamwidth

Beamwidth is defined as the angular separation between two directions on either side of the main lobe where radiation intensity falls to half of its maximum value. This parameter is commonly referred to as half-power beamwidth (HPBW) and plays an important role in determining directional resolution of an antenna. Narrow beamwidth antennas provide higher directivity and improved signal focusing capability, which makes them particularly suitable for radar communication systems requiring accurate target detection and tracking.

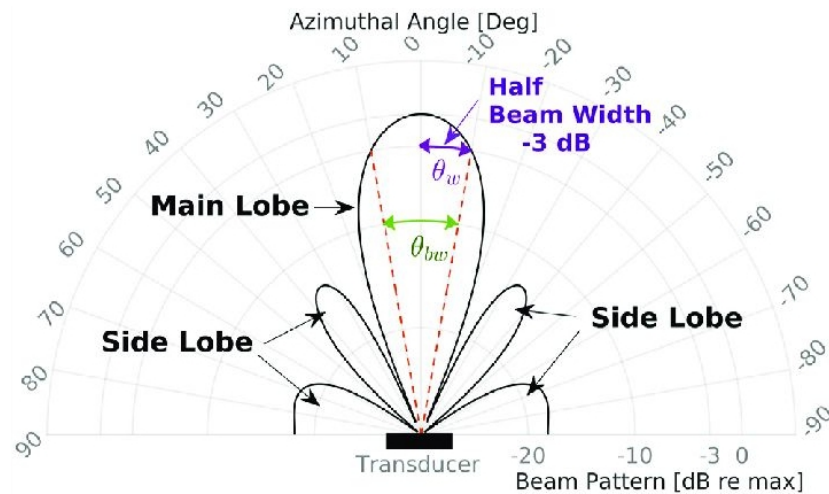


Fig 1.13 Beamwidth

CHAPTER 2

MICROSTRIP ANTENNA

2.1 INTRODUCTION TO MICROSTRIP ANTENNAS

Microstrip antennas are one of the most widely used radiating structures in modern wireless communication systems due to their compact size, lightweight configuration, low fabrication cost, and ease of integration with microwave integrated circuits. A microstrip antenna typically consists of a conducting radiating patch printed on one side of a dielectric substrate with a ground plane on the opposite side. The radiating patch is generally fabricated using copper material, while the substrate is selected based on dielectric constant, thickness, and loss characteristics suitable for the intended operating frequency. Because of their planar geometry and compatibility with printed circuit technology, microstrip antennas are widely used in radar communication systems, satellite communication systems, wireless sensing devices, and high-frequency communication applications. Microstrip antennas operate effectively at microwave and millimeter-wave frequencies, where compact antenna dimensions are required. The ability to modify patch geometry allows designers to improve antenna characteristics such as gain, bandwidth, and impedance matching performance according to application requirements.

2.2 BASIC STRUCTURE OF MICROSTRIP ANTENNA

A microstrip antenna consists of three essential components, namely the radiating patch, dielectric substrate, and ground plane. The radiating patch is responsible for electromagnetic radiation and is usually designed in shapes such as rectangular, circular, triangular, or slot-loaded geometries depending on performance requirements. The dielectric substrate supports the patch mechanically and influences antenna performance parameters such as resonant

frequency, bandwidth, and radiation efficiency. The ground plane reflects electromagnetic energy toward the patch and improves directional radiation characteristics. The dimensions of the radiating patch depend on the operating frequency and dielectric constant of the substrate material. Proper selection of substrate thickness and dielectric constant is important for achieving efficient radiation performance and impedance matching between antenna and transmission line.

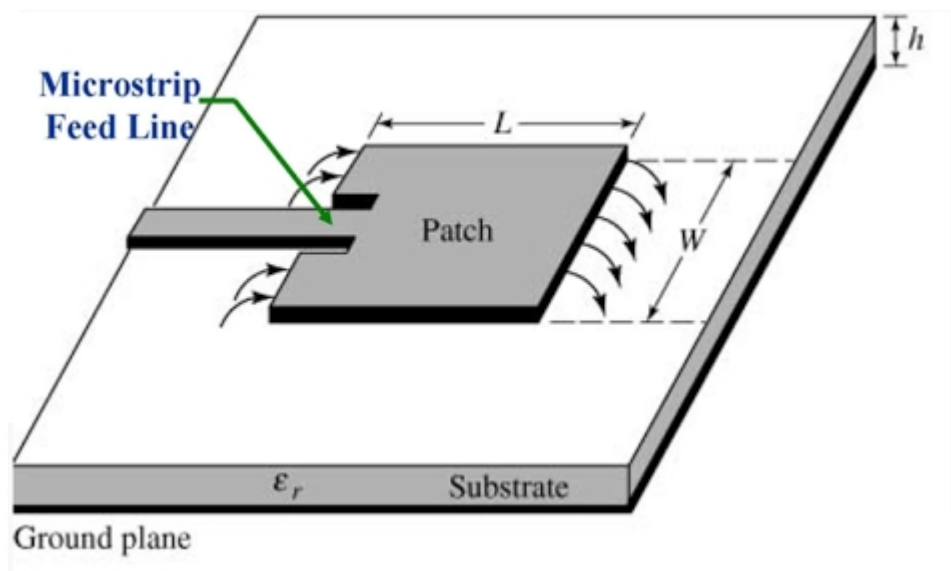


Fig 2.1 Microstrip structure

2.3 WORKING PRINCIPLE OF MICROSTRIP ANTENNA

The working principle of a microstrip antenna is based on fringing field radiation occurring at the edges of the conducting patch. When an RF signal is applied through the feed line, surface currents are induced on the radiating patch. These currents generate electric fields at the edges of the patch that extend beyond the physical boundaries of the conductor into the surrounding air region. These fringing fields are responsible for radiation of electromagnetic waves into free space. The resonant frequency of a microstrip antenna mainly depends on patch dimensions, substrate thickness, and dielectric constant of the substrate

material. Efficient radiation occurs when the patch length is approximately equal to half of the guided wavelength within the dielectric medium.

2.4 MODES OF OPERATION IN MICROSTRIP ANTENNA

Microstrip patch antennas operate primarily in transverse magnetic (TM) modes. Among these modes, the dominant mode of operation for a rectangular patch antenna is the TM_{10} mode. In this mode, the electric field varies sinusoidally along the length of the patch and remains constant along the width. The resonant frequency for the dominant mode is determined by the effective length of the patch conductor. The resonant frequency of the rectangular microstrip antenna operating in dominant mode is expressed as

$$f_r = \frac{c}{2L\sqrt{\epsilon_{eff}}} \quad \text{----- 2.1}$$

Where f_r represents resonant frequency, c represents velocity of light, L represents effective length of the patch, and ϵ_{eff} represents effective dielectric constant of the substrate.

2.5 ADVANTAGES OF MICROSTRIP ANTENNAS

Microstrip antennas provide several advantages compared to conventional antenna structures because of their planar geometry and compatibility with printed circuit fabrication techniques. These antennas have low-profile structure, which allows easy integration with microwave integrated circuits and modern communication devices. Their lightweight configuration makes them suitable for portable wireless systems and airborne communication applications. Microstrip antennas can be fabricated using standard photolithographic techniques, resulting in reduced manufacturing cost and improved reproducibility. In addition, they support both linear and circular polarization and can be easily configured into

array structures to achieve higher gain and improved radiation characteristics required for radar communication systems.

2.6 FEEDING TECHNIQUES OF MICROSTRIP ANTENNAS

Microstrip antennas can be excited using several feeding techniques depending on the design requirements and operating frequency. The most commonly used feeding methods include microstrip line feeding, coaxial probe feeding, aperture coupling, and proximity coupling techniques. In microstrip line feeding, a conducting strip is connected directly to the radiating patch, providing simple fabrication and easy impedance matching. Coaxial probe feeding involves connecting the inner conductor of a coaxial cable directly to the patch while the outer conductor is connected to the ground plane. Aperture coupling uses an aperture slot in the ground plane to transfer electromagnetic energy from the feed line to the patch. Proximity coupling uses an aperture slot in the ground plane to transfer electromagnetic energy from the feed line to the patch.

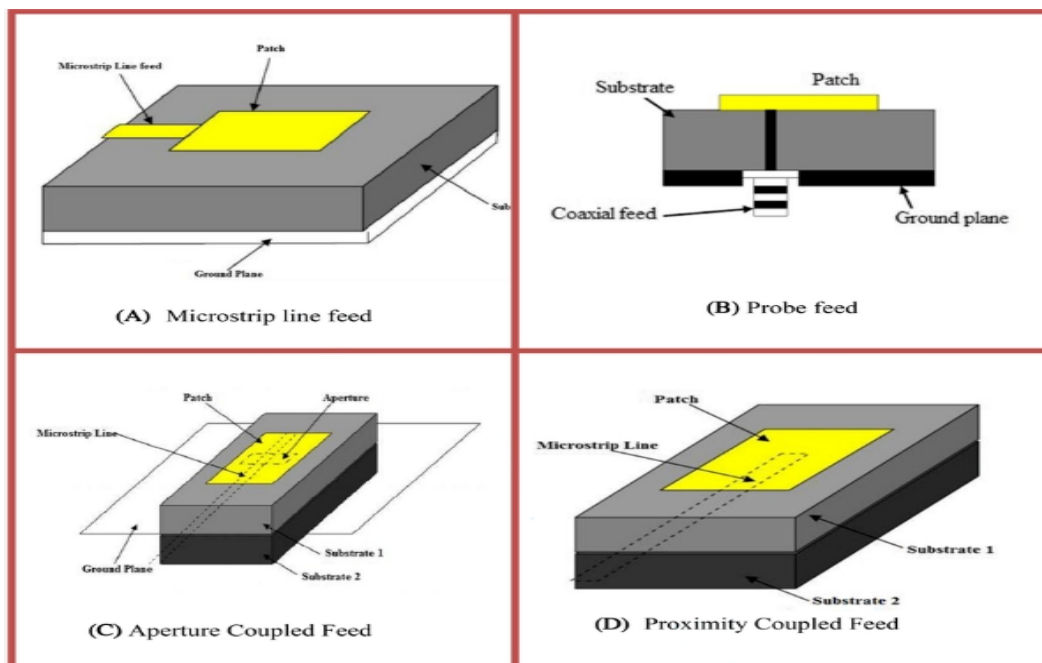


Fig 2.2 Feeding Techniques

2.7 SUBSTRATE MATERIALS USED IN MICROSTRIP ANTENNAS

The performance of a microstrip antenna strongly depends on the properties of the dielectric substrate used in its construction. Important substrate parameters include dielectric constant, thickness, and loss tangent. Substrates with low dielectric constant provide better radiation efficiency and wider bandwidth, whereas substrates with high dielectric constant reduce antenna size but may increase dielectric losses. Commonly used substrate materials include FR-4 epoxy, Rogers RT/Duroid, and Rogers RO-series dielectric materials. For high-frequency radar communication applications, low-loss substrates such as Rogers materials are preferred because they provide improved impedance matching and reduced signal attenuation.

Material	Resonance Frequency	Max U(radiation directivity) *(W/sr)	Peak Directivity	Peak Gain	Peak Realized Gain	Radiation Efficiency	Front to Back Ratio
FR4 eproxy	5.67	0.00103	3.931	1.703	1.530	0.433	236.126
	6.52	0.00027	1.261	0.467	0.407	0.370	9.484
	7.66	0.00036	1.063	0.617	0.534	0.580	49.480
	8.88	0.00116	4.055	2.734	1.721	0.674	100.122
Rogers/RT Duroid 5870	5.67	0.00115	1.980	1.721	1.698	0.869	7.260
	6.52	0.00123	2.019	1.959	1.822	0.971	37.659
	7.66	0.00268	4.089	4.009	3.961	0.980	814.598
	8.88	0.00178	2.786	2.695	2.634	0.967	17.958
Al2O3 Ceramic	5.67	0.00018	0.685	0.676	0.267	0.987	74.667
	6.52	0.00027	1.199	1.208	0.395	1.007	3.314
	7.66	0.00190	3.415	3.457	2.828	1.012	64.310
	8.88	0.00068	1.535	1.793	1.019	1.168	7.885
ArlonDiClad 522	5.67	0.00025	1.020	0.900	0.378	0.882	57.575
	6.52	0.00098	1.923	1.858	1.459	0.967	39.307
	7.66	0.00092	2.127	2.048	1.369	0.962	506.158
	8.88	0.00148	2.782	2.653	2.200	0.954	13.780
Neltec NX9240	5.67	0.00061	1.351	1.165	0.911	0.863	6.032
	6.52	0.00115	1.909	1.846	1.713	0.967	39.379
	7.66	0.00177	2.987	2.892	2.640	0.968	3485.500
	8.88	0.00158	2.598	2.454	2.360	0.944	14.981
Benzocyclobuten	5.67	0.00019	1.096	1.065	0.279	0.971	34.873
	6.52	0.00061	1.351	1.165	0.911	0.863	6.032
	7.66	0.00057	1.805	1.821	0.853	1.009	110.038
	8.88	0.00147	3.201	3.177	2.195	0.992	30.009

Table 2.3 Different types of substrate

2.8 APPLICATIONS OF MICROSTRIP ANTENNAS

Microstrip antennas are widely used in several modern wireless communication systems because of their compact size and ease of fabrication. These antennas are commonly used in radar communication systems, satellite communication systems, mobile communication devices, wireless sensor networks, and medical diagnostic equipment. In radar communication applications, microstrip antennas provide directional radiation characteristics required for accurate object detection and tracking. They are also widely used in aerospace communication systems, navigation systems, and remote sensing applications where compact antenna structures are required. The flexibility of microstrip antenna geometry allows designers to modify patch shapes such as slot-loaded structures and T-shaped configurations to improve antenna performance for specific applications.

CHAPTER 3

LITERATURE SURVEY

3.1 Study and Design of Heart-Shaped Microstrip Patch Antenna for SART Applications (2025)

Pravesh S., Nandana M. K., Akshaya R., and Christina Josephine Malathi A. (2025) presented a compact heart-shaped microstrip patch antenna designed for Search and Rescue Transponder (SART) radar applications operating in the X-band frequency range of 9.2 GHz to 9.5 GHz. The antenna was fabricated on an FR-4 dielectric substrate with relative permittivity 4.4 and substrate thickness 1.6 mm, while the overall antenna dimensions were reported as 40 mm $\times$ 40 mm $\times$ 1.6 mm. The heart-shaped radiating geometry was selected to enhance beamwidth and improve directional radiation performance compared with conventional rectangular patch antennas. Simulation results demonstrated excellent impedance matching with a return loss of -42.2 dB and VSWR of 1.0152, indicating minimal reflection losses at the operating frequency.

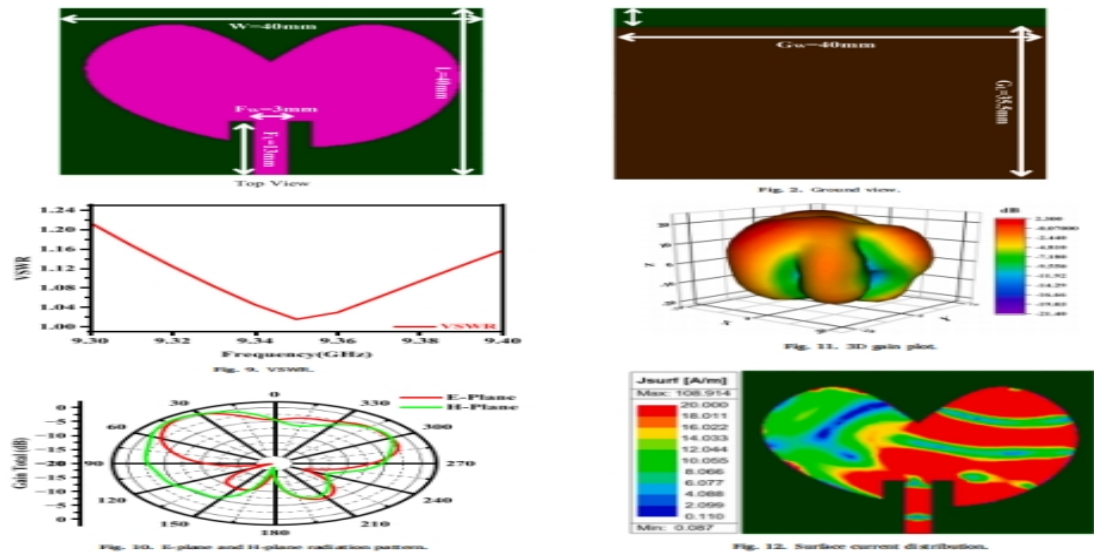


Fig 3.1 Parameter of Antenna for SART Applications

3.2 Design of a Novel Compact and High-Efficiency T-Slot Microstrip Antenna for 28 GHz (2025)

Tariq Iqbal, Muhammad Irfan Khattak, Waqar Hussain, and Taimur Ahmed Khan (2025) proposed a compact T-slot microstrip patch antenna operating at 28 GHz for millimeter-wave wireless communication applications. The antenna was designed using a Rogers RT5880 substrate with dielectric constant 2.2, thickness 0.8 mm, and loss tangent 0.0009, which significantly reduced dielectric losses at high frequencies. The introduction of a T-shaped slot in the radiating patch modified current distribution and improved impedance bandwidth performance. The antenna achieved a radiation efficiency of 81.18%, peak gain of 7.23 dB, and impedance bandwidth covering 27 GHz to 29 GHz, making it suitable for high-frequency communication and radar sensing systems. Comparative analysis with multiple substrate materials confirmed that Rogers RT5880 provided superior electromagnetic performance and reduced surface-wave losses. The slot-loaded configuration enhanced radiation stability and improved antenna directivity for millimeter-wave sensing applications.

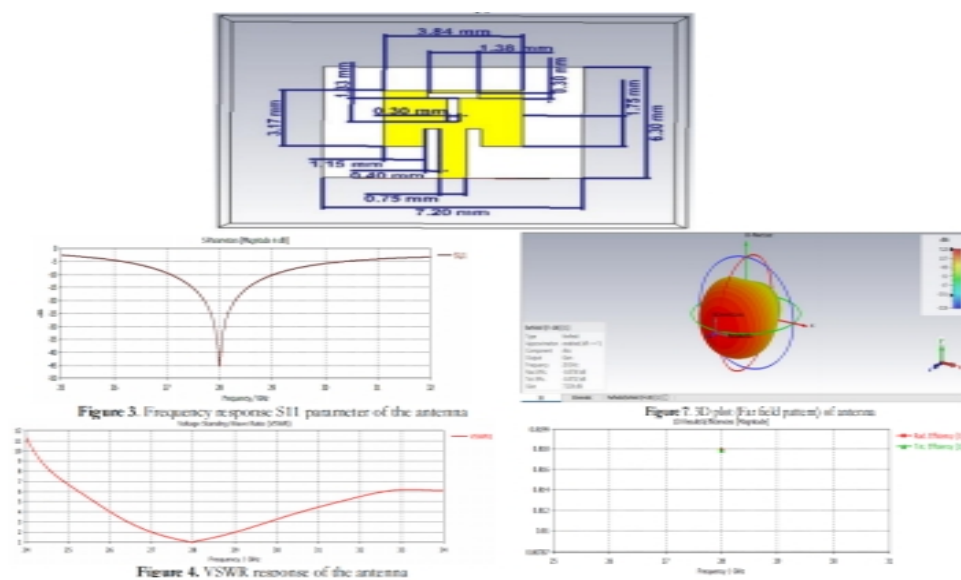


Fig 3.2 Parameter of High-Efficiency T-Slot Microstrip Antenna

3.3 Design and Analysis of Microstrip Antenna Using Double Slot Patch at 28 GHz Frequency for 5G Technology (2025)

Esa Noer Fadhila, Endah Setyowati, and Dewi Indriati Hadi Putri (2025) designed a high-performance double-slot microstrip patch antenna operating at 28 GHz using hybrid T-slot and U-slot geometrical modifications to improve bandwidth and gain performance. The antenna structure was simulated using CST Studio Suite with array configurations of 8×8 and 16×16 radiating elements to enhance radiation characteristics. The optimized antenna array achieved a maximum gain of 18.6 dBi, return loss of -35.95 dB, and VSWR of 1.03, demonstrating excellent impedance matching performance. The bandwidth achieved was approximately 1.44 GHz, which is suitable for millimeter-wave communication and radar sensing applications. The incorporation of slot-loading techniques significantly modified surface current distribution across the patch structure and enabled improved directional radiation characteristics compared with conventional patch antenna configurations.

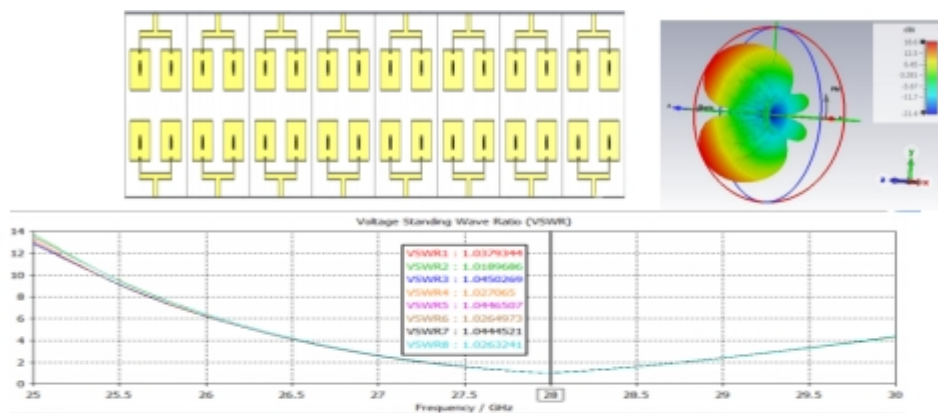


Fig.11. VSWR of Microstrip Antenna 16×16

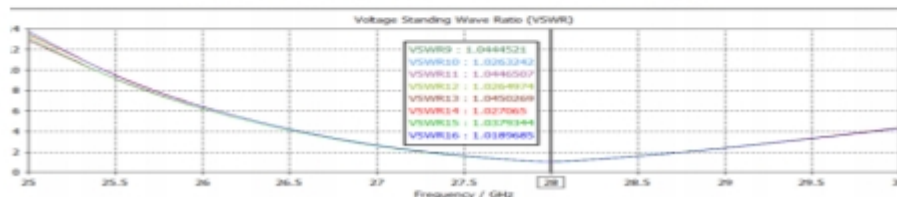


Fig.12. VSWR of Microstrip Antenna 16×16

Fig 3.3 Parameter of Double Slot Patch at 28 GHz Frequency

3.4 On the Design and Analysis of Multiband Microstrip Patch Antenna for Wireless Body Area Network Applications (2025)

D. Gopinath and P. Marichamy (2025) proposed a compact multiband microstrip patch antenna designed for Wireless Body Area Network (WBAN) applications operating across multiple resonant frequencies between 1.99 GHz and 8.07 GHz. The antenna was fabricated on an FR-4 substrate with patch dimensions of $38 \text{ mm} \times 29.4 \text{ mm}$, and multiband operation was achieved using slot-loading techniques applied to the radiating patch geometry. The design produced seven distinct resonant frequency bands through parametric optimization using HFSS electromagnetic simulation software. Performance parameters such as reflection coefficient, gain, and specific absorption rate (SAR) were analyzed to ensure compatibility with wearable biomedical communication systems. The slot-loaded configuration improved impedance matching and enabled compact multiband operation without increasing antenna size significantly, making the antenna suitable for portable healthcare monitoring devices and short-range wireless communication systems.

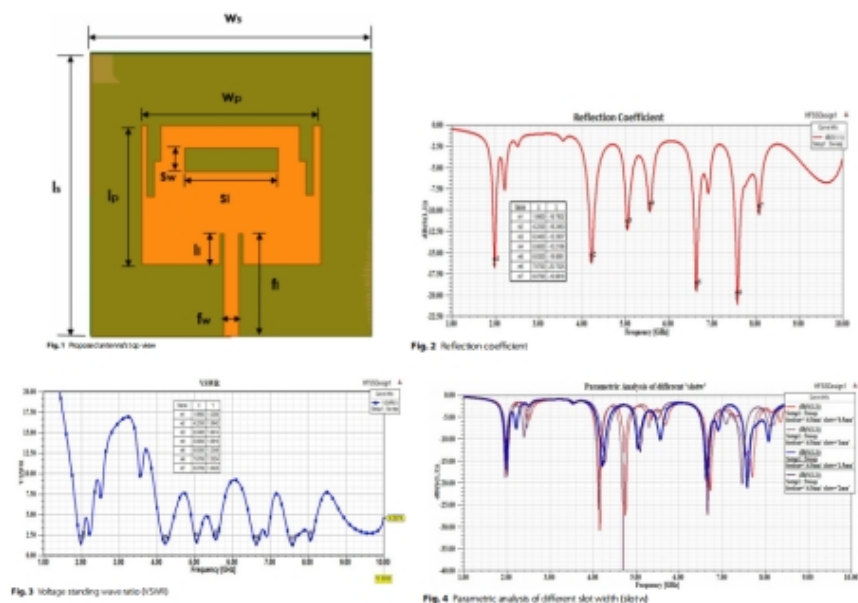


Fig 3.4 Parameter of Multiband Microstrip Patch Antenna

3.5 Microstrip Patch Antenna Design at 10 GHz for X-Band Applications (2024)

Mehmet Karahan, Mertcan Inal, Alperen Dilmen, Furkan Lacinkaya, Ahmet Nuri Akay, and Cosku Kasnakoglu (2024) designed a rectangular microstrip patch antenna operating at 10 GHz for X-band radar and satellite communication applications. The antenna structure was analyzed using HFSS, MATLAB, and AWR simulation tools to evaluate transmission-line parameters and radiation characteristics. The antenna geometry included both top-view and side-view structural layouts for accurate electromagnetic modeling. Simulation results confirmed stable impedance matching performance and directional radiation characteristics in both E-plane and H-plane patterns. The proposed antenna structure demonstrated suitability for applications such as air-traffic control radar systems, weather monitoring radar systems, defense tracking systems, and vehicle speed detection radar systems due to its reliable performance within the X-band frequency spectrum.

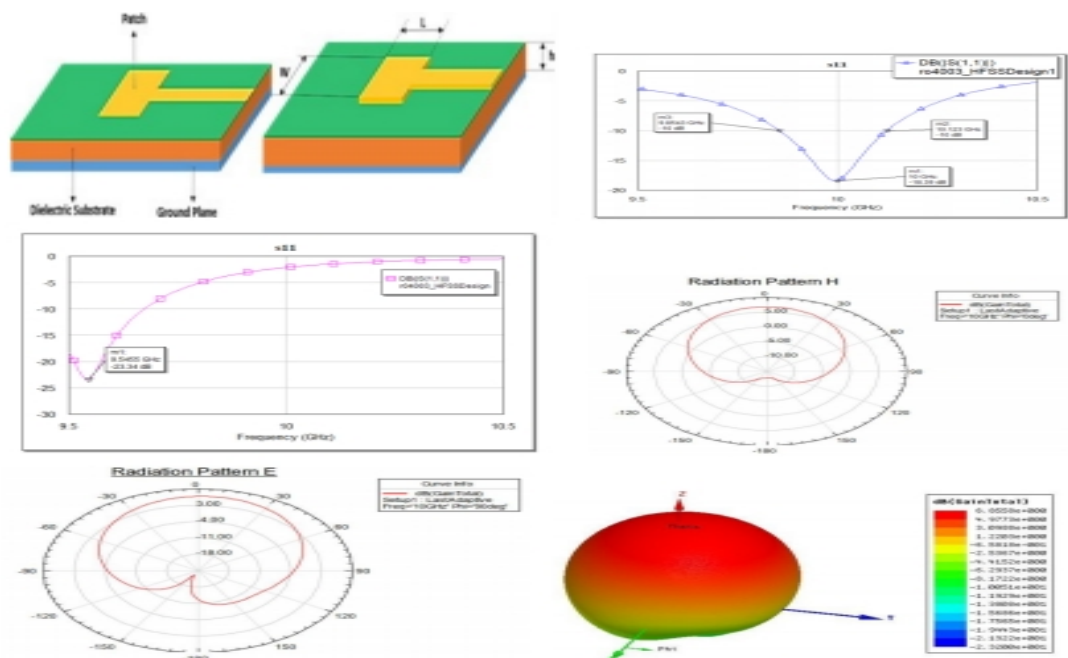


Fig 3.5 Parameter of X-Band Applications

3.6 A Multiband T-Shaped Antenna Array for 6G Mobile Communication (2024)

Sunday Achimugu, Abraham Usman Usman, Suleiman Zubair, Michael David, Abdulkadir Olayinka Abdulbaki, and Hassan Musa Abdullahi (2024) proposed a multiband double T-shaped microstrip antenna array designed for broadband wireless communication systems operating within the frequency range of 4 GHz to 16 GHz. The antenna was fabricated using a Rogers RT5880 substrate with dielectric constant 2.2 and thickness approximately 0.766 mm, while the overall antenna structure dimensions were reported as $30.9 \text{ mm} \times 26.63 \text{ mm} \times 0.1 \text{ mm}^3$. The antenna incorporated a fractal T-slot configuration at the rear side of the microstrip radiator to enhance impedance matching and radiation characteristics. Simulation results showed that the antenna achieved a peak gain of 7.99 dBi with 92% radiation efficiency, and maintained reflection coefficient values below -10 dB across multiple operating frequency bands including 6.62–7.54 GHz, 8.27–8.78 GHz, 10.98–11.78 GHz, and 13.65–15.4 GHz. The multiband operation and high radiation efficiency demonstrated the suitability of the antenna for next-generation broadband communication and microwave radar sensing applications requiring compact and high-performance antenna arrays.

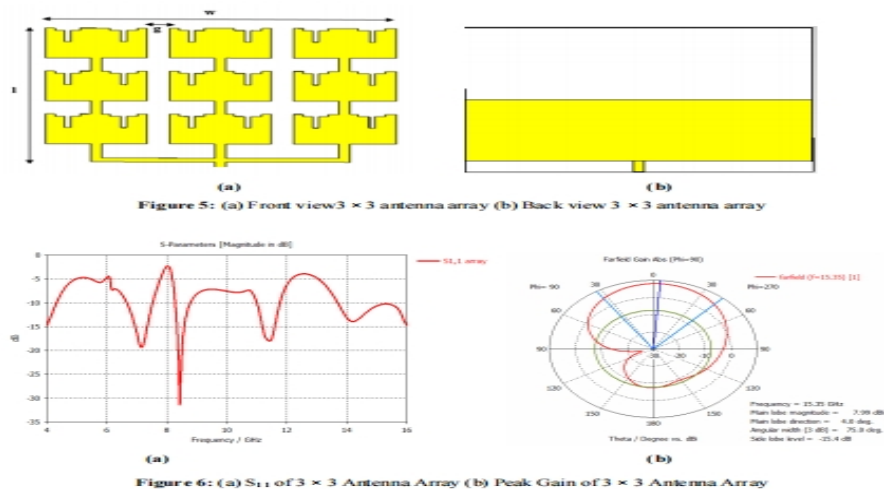


Fig 3.6 Parameter of T-Shaped Antenna Array

3.7 Design of an Improved Microstrip Antenna Operating at a Frequency Band of 28 GHz (2024)

S. O. Zakariyya, V. O. Omole, B. O. Sadiq, R. A. Alao, J. A. Adesina, and E. Obi (2024) developed an improved microstrip patch antenna operating at 28 GHz using a Rogers RT5880 LZ substrate having dielectric constant 1.96 and substrate thickness 0.762 mm. The antenna employed a quarter-wave transformer feeding technique to achieve impedance matching with a standard 50-ohm transmission line, while a U-shaped slot modification in the ground plane was introduced to enhance radiation performance. Simulation results demonstrated a return loss of -21.4 dB, gain of 8.19 dB, bandwidth of 2.026 GHz, and VSWR of 1.18, confirming improved impedance matching and radiation stability compared with conventional rectangular microstrip antennas. The proposed antenna structure effectively addressed limitations such as narrow bandwidth and moderate gain typically observed in high-frequency patch antennas, making it suitable for millimeter-wave radar communication and 5G sensing applications.

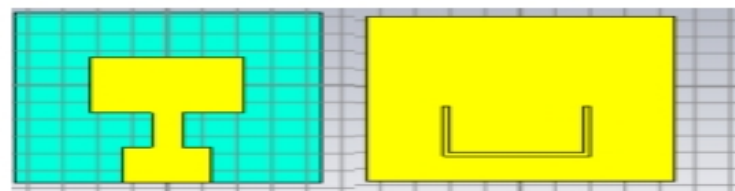


Figure 1: Front and back view of the designed antenna

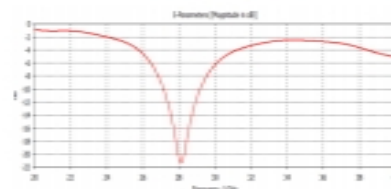


Figure 2: Return loss plot

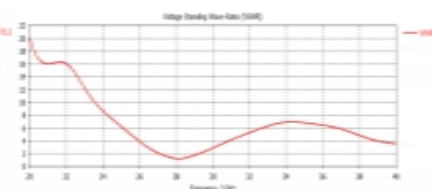


Figure 3: voltage standing wave ratio

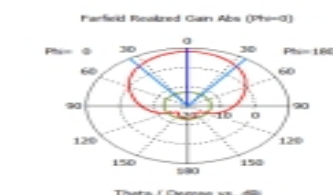


Figure 4a: 2D Radiation pattern plot

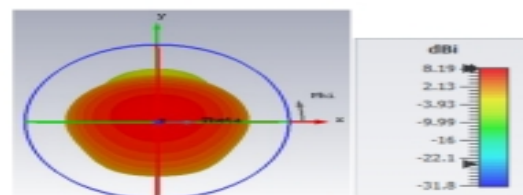


Figure 4b: 3D Radiation pattern plot

Fig 3.7 Parameter of Improved Microstrip Antenna of 28 GHz

3.8 A High-Gain Dual-Band Slotted Microstrip Patch Antenna for 5G Cellular Mobile Phones (2024)

Mouaaz Nahas (2024) proposed a compact dual-band slotted microstrip patch antenna operating at 26 GHz and 28 GHz using a hammer-slot geometry to enhance gain and radiation efficiency. The antenna structure incorporated multiple slot elements within the rectangular patch radiator to modify surface current distribution and improve directional radiation performance. Simulation results demonstrated significant gain enhancement compared with conventional rectangular patch antennas operating in the same millimeter-wave frequency band. The optimized slot configuration improved impedance bandwidth and radiation stability while maintaining compact antenna dimensions suitable for integration into modern wireless communication devices. The antenna design was found particularly suitable for millimeter-wave radar sensing systems and high-frequency wireless communication applications due to its improved gain performance and dual-band operational capability.

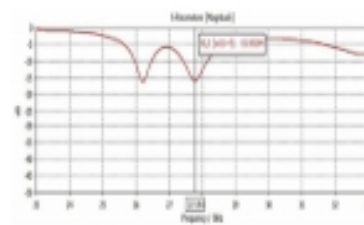
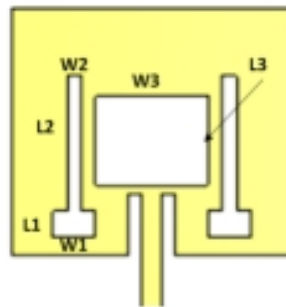


Fig. 2. S_{11} as a function of frequency for the proposed antenna.

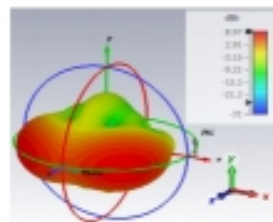


Fig. 3. The radiation pattern for the hammer-slotted antenna at 26 GHz.

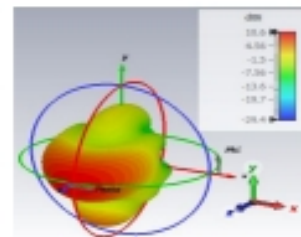


Fig. 4. The radiation pattern for the hammer-slotted antenna at 28 GHz.

Fig 3.8 Parameter of High-Gain Dual-Band Slotted Microstrip Patch Antenna

3.9 A Dual-Band T-Slot Microstrip Patch Antenna for Wireless Communication and Radar Application (2023)

Stella Ifeoma Orakwue and Praise Igochi Onu (2023) designed a dual-band T-slot microstrip patch antenna operating at 5.923 GHz and 7.444 GHz for wireless communication and radar sensing applications. The antenna geometry incorporated a T-shaped slot in the radiating patch structure to improve impedance bandwidth and radiation characteristics compared with conventional patch antenna configurations. The antenna was simulated using CST electromagnetic simulation software and achieved return loss values of -33.696 dB and -18.464 dB, along with gain values of 5.45 dB and 4.158 dB at the respective operating frequencies. The slot-loaded configuration altered surface current distribution across the radiating patch and enabled stable dual-band operation without increasing antenna dimensions significantly. The proposed antenna structure demonstrated suitability for microwave radar communication and WLAN-based sensing applications due to its compact size and improved impedance matching characteristics.

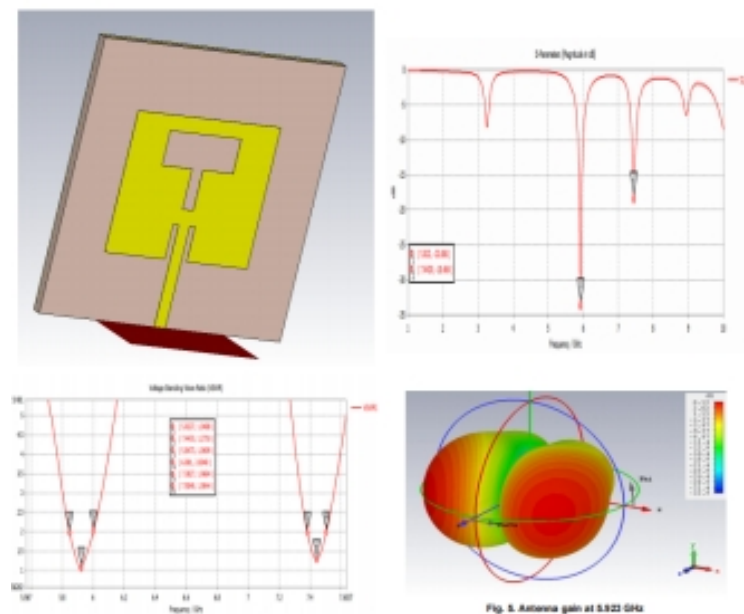


Fig 3.9 Parameter of Dual-Band T-Slot Microstrip Patch Antenna

3.10 Microstrip Antenna Development for Radar Sensor (2023)

Lajos Nagy (2023) developed a compact microstrip antenna designed for radar-based level measurement sensor applications operating at a center frequency of approximately 25 GHz. The antenna structure incorporated a circular radiating patch geometry combined with an optimized feeding network and an additional ring structure surrounding the antenna array to suppress sidelobe levels and improve directional radiation characteristics. The antenna was implemented using a low-loss Rogers RT5880 substrate, which contributed to improved radiation efficiency and reduced dielectric losses at high frequencies. The proposed antenna achieved approximately 1 GHz bandwidth with stable gain performance and reduced sidelobe levels, making it suitable for non-contact radar sensing applications in industrial environments such as chemical processing plants, storage tanks, and automated monitoring systems. The compact structure and high-frequency operation demonstrated the effectiveness of microstrip antenna technology for precision radar measurement applications.

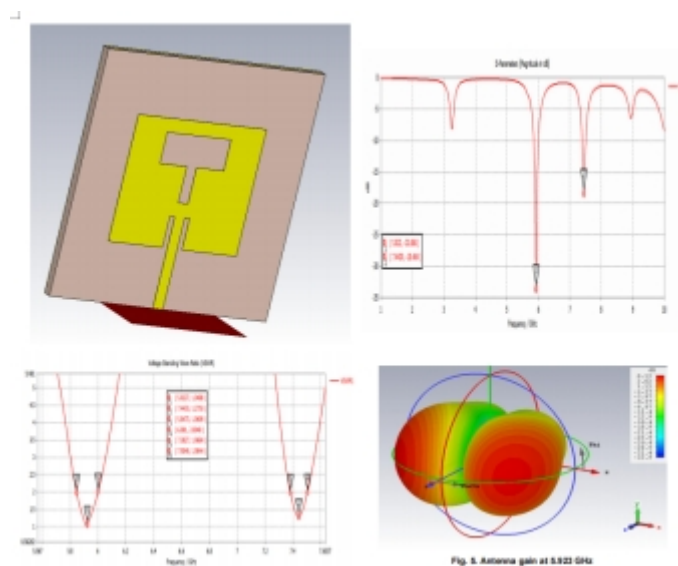


Fig 3.10 Parameter of Microstrip Antenna Development

CHAPTER 4

PROPOSED SYSTEM

4.1 Objective

The objective of the proposed work is to design and analyze a compact T-shaped microstrip patch antenna operating at 26 GHz for radar communication applications. Modern radar systems require antennas with small physical dimensions, low return loss, stable radiation characteristics, and efficient impedance matching. Conventional rectangular microstrip patch antennas suffer from narrow bandwidth and moderate radiation efficiency at microwave frequencies. Therefore, a modified T-shaped radiating patch structure is proposed to improve antenna performance. In addition to patch modification, a defected ground structure (DGS) is introduced in the ground plane to enhance impedance matching and reduce surface wave losses. The antenna is designed using Rogers RO4003 substrate material due to its low dielectric loss and stable electrical properties at high frequencies. The structure is modeled and analyzed using ANSYS HFSS simulation software to evaluate important antenna parameters such as return loss, VSWR, radiation pattern, and gain suitable for radar communication systems operating in the K-band frequency region.

4.2 T Shaped Antenna

A T-shaped microstrip patch antenna is a modified version of the conventional rectangular microstrip patch antenna in which additional rectangular extensions are introduced to form a T-shaped radiating structure. This modification increases the effective electrical length of the antenna without significantly increasing its overall physical size. As a result, the antenna can operate efficiently at higher frequencies while maintaining compact dimensions,

which makes it suitable for modern microwave and radar communication systems.

The structure of a T-shaped microstrip antenna typically consists of two main parts known as the horizontal arm and the vertical arm. These arms are formed by combining rectangular conducting sections printed on the top surface of a dielectric substrate, while a conducting ground plane is placed on the bottom surface. The horizontal arm improves the distribution of surface current across the width of the patch, whereas the vertical arm increases the current path length along the length of the patch. This combined effect enhances the impedance matching between the feed structure and the radiating patch compared to a conventional rectangular patch antenna.

Another important advantage of the T-shaped microstrip patch antenna is its ability to improve return loss characteristics by reducing signal reflection at the input terminal. The modified geometry allows better control over the resonant frequency and provides flexibility in antenna design without increasing substrate size. In addition, the improved surface current distribution results in enhanced radiation efficiency and stable directional radiation characteristics.

The compact structure of the T-shaped microstrip antenna makes it suitable for applications where space is limited and efficient performance at microwave frequencies is required. Because of these advantages, T-shaped microstrip patch antennas are widely used in radar systems, wireless communication devices, satellite communication systems, and other high-frequency sensing applications where compact size and reliable radiation performance are essential.

4.3 Design Calculation of the Proposed Antenna

The dimensions of the proposed microstrip patch antenna are calculated using transmission line model equations. The antenna is designed to operate at a resonant frequency of 26 GHz using Rogers RO4003 substrate having dielectric constant $\epsilon_r=3.55$ and substrate thickness $h=1.6$ mm.

Calculation of Patch Width (W)

The width of the microstrip patch antenna is calculated using

$$W = \frac{c}{2f_r} \sqrt{\frac{2}{\epsilon_r + 1}} \quad \text{----- 4.1}$$

Substituting values:

$$W = \frac{3 \times 10^8}{2 \times 26 \times 10^9} \sqrt{\frac{2}{3.55 + 1}}$$
$$W = 3.82 \text{ mm}$$

Calculation of Effective Dielectric Constant (ϵ_{eff})

The effective dielectric constant is calculated using

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + 12 \frac{h}{W} \right)^{-1/2} \quad \text{----- 4.2}$$

Substituting values:

$$\epsilon_{eff} = \frac{3.55 + 1}{2} + \frac{3.55 - 1}{2} \left(1 + 12 \frac{1.6}{3.82} \right)^{-1/2}$$
$$\epsilon_{eff} = 2.80$$

Calculation of Effective Length (L_{eff})

The effective patch length is calculated as

$$L_{eff} = \frac{c}{2f_r \sqrt{\epsilon_{eff}}} \quad \text{----- 4.3}$$

Substituting values:

$$L_{eff} = \frac{3 \times 10^8}{2 \times 26 \times 10^9 \sqrt{2.80}}$$

$$L_{eff} = 3.45 \text{ mm}$$

Calculation of Length Extension ΔL

The extension length due to fringing fields is

$$\Delta L = 0.412h \frac{(\epsilon_{eff} - 0.258)(W/h + 0.8)}{(\epsilon_{eff} + 0.3)(W/h + 0.264)} \quad \text{----- 4.4}$$

Substituting values:

$$\Delta L = 0.412(1.6) \frac{(2.80 - 0.258)(3.82/1.6 + 0.8)}{(2.80 + 0.3)(3.82/1.6 + 0.264)}$$

$$\Delta L = 0.669 \text{ mm}$$

Calculation of Actual Patch Length (L)

The actual patch length is calculated as

$$L = L_{eff} - 2\Delta L \quad \text{----- 4.5}$$

Substituting values:

$$L = 3.45 - 2(0.669)$$

$$L = 2.11 \text{ mm}$$

Ground Plane Dimensions

Ground plane dimensions are calculated using

$$L_g = L + 6h \quad \text{----- 4.6}$$

$$L_g = 2.11 + 6(1.6)$$

$$L_g = 11.71 \text{ mm}$$

$$W_g = W + 6h \quad \text{----- 4.7}$$

$$W_g = 3.82 + 6(1.6)$$

$$W_g = 13.42 \text{ mm}$$

4.4 Antenna Structure Layout

The proposed antenna structure consists of three main layers, namely the radiating patch, dielectric substrate, and ground plane. The radiating patch is designed in a T-shaped configuration and is printed on the top surface of the dielectric substrate. The substrate selected for this design is Rogers RO4003, which provides stable electrical performance at microwave frequencies due to its low dielectric loss and moderate dielectric constant. The use of Rogers

RO4003 improves radiation efficiency and minimizes signal attenuation, making it suitable for high-frequency radar communication applications.

The overall dimensions of the substrate are selected as 20 mm in length and 8 mm in width with a thickness of 1.6 mm. These dimensions ensure compact antenna size while maintaining proper electromagnetic wave propagation characteristics at the operating frequency of 26 GHz. A full ground plane of the same length and width as the substrate is placed on the bottom surface, which acts as a reference conductor and helps improve radiation directivity by reducing backward radiation.

The radiating patch is formed using two rectangular sections combined to produce a T-shaped geometry. The horizontal arm of the patch is designed with a length of 11.4 mm and a width of 2.5 mm, while the vertical arm has a length of 3.8 mm and a width of 4 mm. This modified geometry increases the effective current path length compared to a conventional rectangular patch antenna. As a result, impedance matching improves and return loss is reduced at the desired operating frequency. The T-shaped configuration also enhances surface current distribution along the patch edges, which contributes to improved radiation performance.

To further enhance antenna performance, a square-shaped defected ground structure is introduced at the center of the ground plane. The presence of this slot modifies the current distribution in the ground plane and suppresses unwanted surface wave propagation. This improves impedance bandwidth and reduces reflection losses, thereby enhancing the overall efficiency of the antenna. The defected ground structure also contributes to compact antenna design by allowing better resonance control without increasing antenna dimensions.

The antenna is excited using a lumped port feeding technique positioned between the radiating patch and the ground plane. The selected feed arrangement ensures efficient transfer of electromagnetic energy from the source to the radiating structure and improves impedance matching at the operating frequency. A radiation boundary is defined around the antenna structure during simulation to replicate free-space propagation conditions and to obtain accurate far-field radiation characteristics. The combination of T-shaped patch geometry, Rogers RO4003 substrate, and defected ground structure results in a compact antenna design suitable for radar communication applications operating in the 26 GHz frequency band.

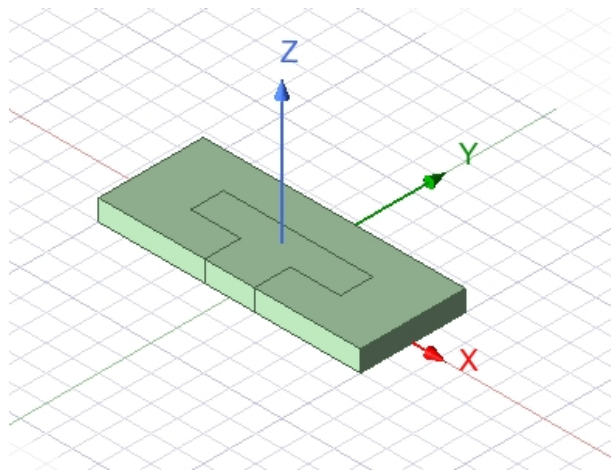


Fig 4.1 Antenna Structure

4.5 Simulation Tool

The proposed antenna structure is designed and analyzed using ANSYS High Frequency Structure Simulator (HFSS), which is a three-dimensional electromagnetic simulation software widely used for modeling microwave and antenna structures. HFSS is based on the finite element method (FEM), which provides accurate numerical solutions for complex electromagnetic field

problems. It allows precise analysis of antenna performance by solving Maxwell's equations over the entire simulation domain.

In this work, the geometry of the T-shaped microstrip antenna is created using the HFSS modeler environment, and the Rogers RO4003 substrate material properties are assigned to the dielectric layer. The radiating patch and ground plane are defined using perfect electric conductor boundary conditions. A lumped port excitation is applied between the feed structure and the ground plane to provide input signal excitation for the antenna. Radiation boundary conditions are assigned around the antenna structure to simulate free-space electromagnetic wave propagation. A frequency sweep is performed around the operating frequency of 26 GHz to determine the resonant characteristics of the antenna. HFSS is used to obtain important antenna parameters such as return loss, voltage standing wave ratio (VSWR), radiation pattern, and gain, which are essential for evaluating antenna performance in radar communication applications.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Structure in CAD

The three-dimensional structure of the proposed antenna is designed using ANSYS HFSS electromagnetic simulation software. The antenna consists of a T-shaped radiating patch printed on the top surface of a Rogers RO4003 dielectric substrate and a full ground plane printed on the bottom surface. The substrate dimensions are selected as $20 \text{ mm} \times 8 \text{ mm} \times 1.6 \text{ mm}$ to ensure compact antenna operation at the operating frequency of 26 GHz.

The radiating patch is formed by combining two rectangular sections that create a T-shaped geometry. The horizontal arm of the patch has dimensions $11.4 \text{ mm} \times 2.5 \text{ mm}$, while the vertical arm has dimensions $3.8 \text{ mm} \times 4 \text{ mm}$. This configuration increases the effective electrical length of the antenna and improves impedance matching characteristics compared to conventional rectangular patch antennas.

A square defected ground structure of size $1 \text{ mm} \times 1 \text{ mm}$ is introduced at the center of the ground plane to suppress surface wave propagation and enhance antenna performance. The antenna is excited using a lumped port feeding technique with a feed width of 3.8 mm. A radiation boundary is defined around the antenna structure to simulate free-space propagation conditions during electromagnetic analysis.



Return loss is an important parameter used to evaluate antenna impedance matching performance. It represents the amount of reflected power from the antenna input terminal. For efficient antenna operation, the return loss should be less than -10 dB at the desired operating frequency.

The improved return loss performance is mainly achieved due to the T-shaped patch configuration and the introduction of the defected ground structure, which modifies the surface current distribution and enhances impedance matching characteristics.

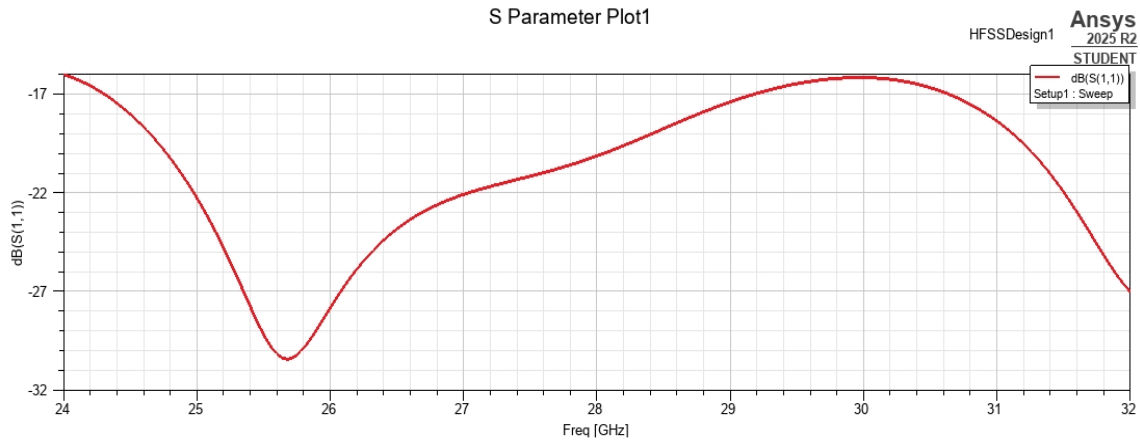


Fig 5.2 Return Loss S11

5.3 Radiation Pattern (RP)

The radiation pattern describes the variation of radiated electromagnetic energy in different directions in space. It is an important parameter used to determine antenna directivity and suitability for radar communication applications. The simulated radiation pattern of the proposed antenna shows directional radiation characteristics with maximum radiation along the broadside direction. This type of radiation behavior is desirable for radar communication systems where focused signal transmission toward a particular direction is required. The directional radiation pattern confirms that the proposed antenna provides stable radiation performance at the operating frequency of 26 GHz. The symmetry observed in the radiation characteristics indicates proper surface current distribution across the radiating patch structure. The obtained radiation behavior demonstrates the suitability of the antenna for microwave radar communication applications requiring directional signal transmission.

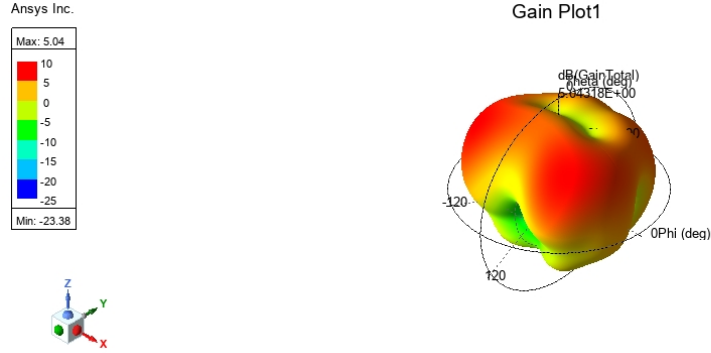


Fig 5.3 3D Gain Plot

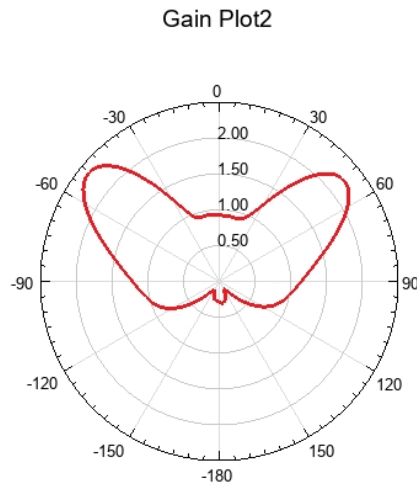


Fig 5.4 2D Gain Plot

5.4 Voltage Standing Wave Ratio (VSWR)

Voltage Standing Wave Ratio (VSWR) is another important parameter used to evaluate impedance matching between the antenna and transmission line. A VSWR value close to unity indicates efficient antenna operation and minimal signal reflection. The simulated VSWR of the proposed antenna is approximately 1.06 at the operating frequency of 26 GHz. This value is very close to unity and confirms excellent impedance matching between the feed structure and the antenna. The low VSWR ensures maximum transfer of input power from the source to the antenna and minimizes signal losses due to reflection.

The improved VSWR performance is achieved due to the optimized T-shaped patch geometry and the introduction of the defected ground structure, which enhances the impedance characteristics of the antenna.

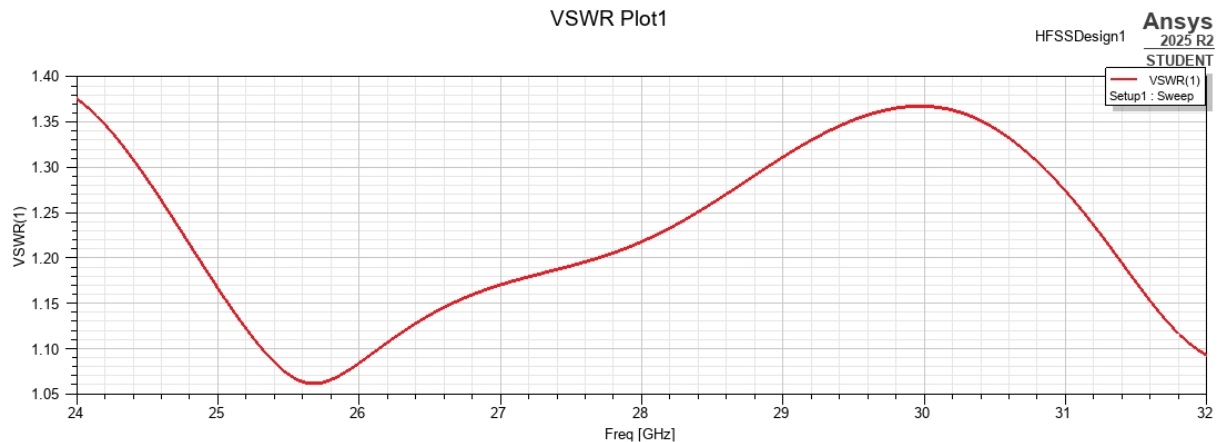


Fig 5.5 VSWR

5.5 Parametric Analysis

Parametric analysis is performed to study the effect of feed width variation on the return loss characteristics of the proposed T-shaped microstrip antenna. The feed width plays an important role in determining impedance matching between the transmission line and the radiating patch. Therefore, optimization of feed width is necessary to achieve resonance at the desired operating frequency of 26 GHz.

In this work, the feed width is varied from 11.4 mm to 11.9 mm while keeping all other antenna parameters constant. The return loss characteristics obtained for different feed width values are analyzed using ANSYS HFSS simulation software. It is observed from the parametric analysis that the resonant frequency shifts slightly with variation in feed width, and impedance matching improves significantly for certain optimized values. When the feed width is selected as 11.4 mm, the antenna resonates near 26 GHz with a return loss close to -30 dB. As the feed width increases gradually to 11.5 mm and 11.6 mm, the resonance behavior slightly shifts and the return loss reduces marginally. Further

increase in feed width beyond this value results in degradation of impedance matching and increase in reflected power.

Among the different values considered during parametric optimization, the feed width of 11.4 mm provides the best return loss performance near the desired operating frequency. Therefore, this optimized feed width is selected for the final antenna structure to achieve efficient impedance matching and stable radiation characteristics.

The parametric study confirms that proper selection of feed width significantly influences antenna resonance and return loss characteristics. Hence, optimization of feed dimensions is essential for obtaining improved antenna performance at microwave radar communication frequencies.

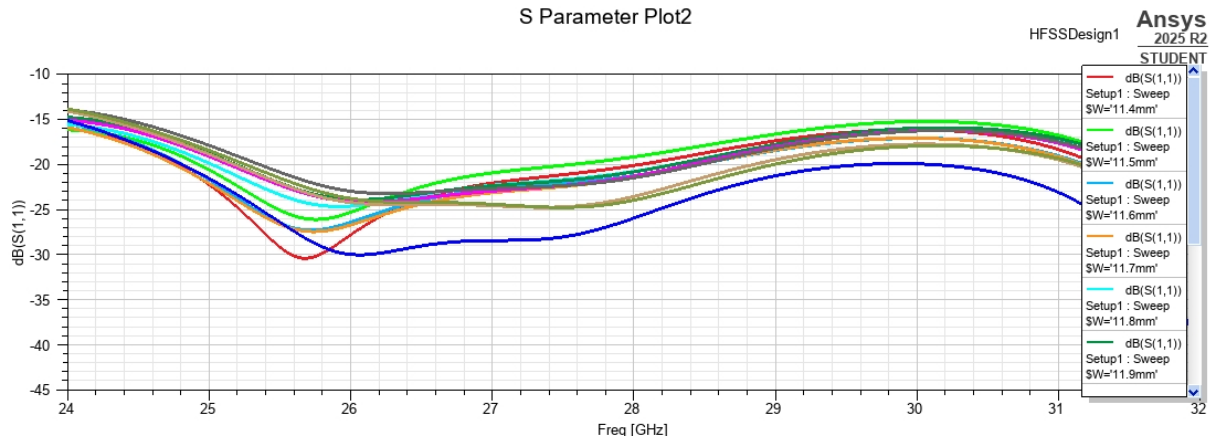


Fig 5.6 Parametric Analysis

5.6 Comparison Table

The performance of the proposed T-shaped microstrip antenna is compared with recently reported antennas operating in microwave and millimeter-wave frequency ranges. The comparison is carried out based on parameters such as operating frequency, antenna geometry, substrate material, gain, and return loss performance. From the comparison results, it is observed

that the proposed antenna provides improved impedance matching with compact dimensions and stable radiation characteristics suitable for radar communication applications at 26 GHz.

Reference	Frequency	Structure	Substrate	Return Loss	Gain
Heart-shaped patch antenna (2025)	9.2–9.5 GHz	Heart-shaped patch	RT5880	−42.2 dB	6.8 dB
T-slot microstrip antenna (2025)	28 GHz	T-slot patch	RT5880	−24 dB	7.23 dB
Double slot hybrid antenna (2025)	28 GHz	T + U slot	RT5880	−28 dB	18.6 dBi
Multiband slot antenna (2025)	1.99–8.07 GHz	Slot-loaded patch	FR-4	−20 dB	5.2 dB
X-band tracking antenna (2024)	10 GHz	Rectangular patch	FR-4	−18 dB	6.1 dB
Double T-shaped antenna array (2024)	4–16 GHz	Double T-patch	RT5880	−25 dB	7.99 dB
Ground-modified patch antenna (2024)	28 GHz	DGS patch	RT5880	−21.4 dB	8.19 dB
Proposed T-shaped microstrip antenna with DGS (This Work)	26 GHz	T-shaped patch antenna with defected ground structure	Rogers RO4003	−30 dB	≈ 5 dB

Table 5.1 Comparasion Table

CHAPTER 6

CONCLUSION

In this work, a compact T-shaped microstrip patch antenna operating at 26 GHz has been successfully designed and analyzed for radar communication applications. The antenna is developed using Rogers RO4003 dielectric substrate with relative permittivity 3.55 and thickness 1.6 mm to achieve stable performance at microwave frequencies. The proposed antenna structure incorporates a modified T-shaped radiating patch along with a defected ground structure (DGS), which improves impedance matching and enhances radiation characteristics without increasing the overall antenna size.

The antenna design is modeled and simulated using ANSYS HFSS electromagnetic simulation software. From the simulation results, the proposed antenna achieves a return loss of approximately -30 dB at the operating frequency of 26 GHz, indicating excellent impedance matching between the feed structure and the radiating patch. The simulated voltage standing wave ratio (VSWR) is close to unity, confirming efficient power transfer from the source to the antenna. The antenna also provides directional radiation characteristics with a gain of approximately 5 dB, which is suitable for radar communication applications in the K-band frequency region.

The incorporation of the T-shaped patch geometry increases the effective electrical length of the antenna and improves surface current distribution across the radiating structure. In addition, the introduction of a square defected ground structure reduces surface wave propagation and enhances antenna efficiency. These improvements contribute to better return loss performance and stable radiation behavior at the desired operating frequency.

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